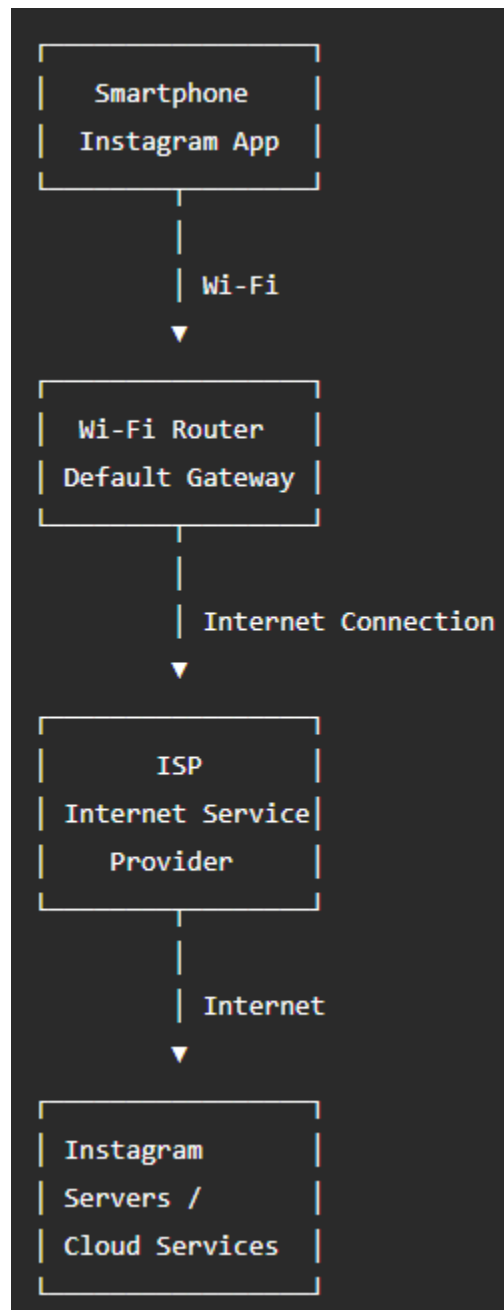


Task 1: Smartphone to Instagram Network Diagram



Task 2: IP Address and Default Gateway

On Windows, I opened Command Prompt and used:

- `ipconfig`

```
C:\Users\panch>ipconfig
```

Windows IP Configuration

Ethernet adapter Ethernet:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Local Area Connection* 2:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

Ethernet adapter VMware Network Adapter VMnet1:

```
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::a248:c4cd:2a41:8efd%8
IPv4 Address. . . . . : 192.168.186.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
```

Ethernet adapter VMware Network Adapter VMnet8:

```
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::6c21:21cf:7f01:6f3%2
IPv4 Address. . . . . : 192.168.182.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2409:40c1:4027:22d4:319a:c6b:29
Temporary IPv6 Address. . . . . : 2409:40c1:4027:22d4:e8fc:195d:f
Link-local IPv6 Address . . . . . : fe80::da02:197f:18c5:84f9%14
IPv4 Address. . . . . : 10.41.112.162
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::64e0:fdff:fe30:47e1%14
                             10.41.112.126
```

Ethernet adapter Bluetooth Network Connection:

```
Media State . . . . . : Media disconnected
```

Meaning

- IP Address:
 - The IP address uniquely identifies my device on the local network and allows network traffic to be delivered to it.
- Default Gateway:
 - The default gateway is usually the IP address of the router. It is used to send traffic from my local network to other networks, including the Internet.

Task 3: Zomato Food Order Journey

Step 1: Device connects to the network

Step 2: DNS lookup

Step 3: Establishing a TCP connection

Step 4: Secure web communication

Step 5: Search for a restaurant

Task 4: Ping www.spotify.com

I opened Command Prompt and ran:

- ping www.spotify.com

Meaning of the Results

- Reply received:
 - Indicates that the destination responded to the ICMP echo request.
- Time:
 - Shows the approximate round-trip time for the packet. Lower values generally indicate lower network latency.
- Packet loss:
 - Shows how many packets did not receive a response. Packet loss can indicate connectivity problems, congestion, filtering, or other network issues.
- TTL:
 - The Time To Live value limits how many network hops a packet can make before being discarded.

- Troubleshooting
 - If all packets are received with 0% packet loss, basic IP connectivity to the destination is working.